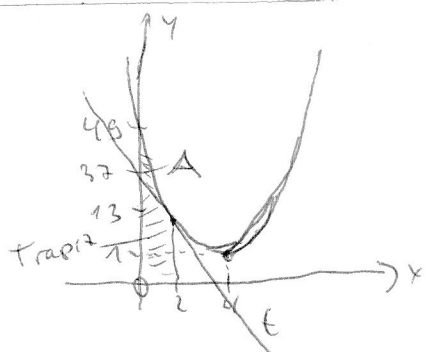


1. a)



b) $f'(x) = 6(x-4)$

$f'(2) = -12$

$t: y = -12(x-2) + 13$
 $= -12x + 37$

$A = \int_0^2 f(x) dx - \frac{37+13}{2} \cdot 2$
 $= \left[(x-4)^3 + x \right]_0^2 - 50 = -6 + 64 - 50 = 8$

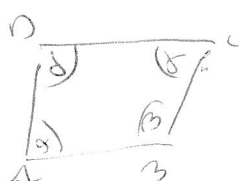
c) $F(x) = (x-4)^3 + x + C$
 $F(1) = -1$
 $-27 + 1 + C = -1$
 $C = 25$

$F''(x) = f'(x) = 6(x-4) = 0$
 $x=4 \quad y=29$
 $F(x) = f(x): f(4) = 1$
 $s: y = 1 \cdot (x-4) + 29$

2. a) $E_{DAB}: \vec{x} = \vec{r}_A + s\vec{AB} + t\vec{AD} = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} + s \begin{pmatrix} -9 \\ 1 \\ -4 \end{pmatrix} + t \begin{pmatrix} 3 \\ -4 \\ -6 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ -6 \end{pmatrix}$
 $s=t=1, C \in E_{ABD}$

$\vec{AB} \times \vec{AD} = \begin{pmatrix} -22 \\ -66 \\ 33 \end{pmatrix} \sim \begin{pmatrix} 2 \\ 6 \\ -3 \end{pmatrix} = \vec{n}_{ABD} = \vec{u}$
 ~~$\vec{u} = \begin{pmatrix} -22 \\ -66 \\ 33 \end{pmatrix}$~~
 senkrecht E_{ABD}

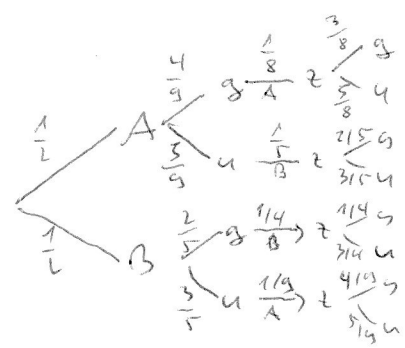
b) $g: \vec{x} = \begin{pmatrix} 8 \\ 11 \\ 6 \end{pmatrix} + m \begin{pmatrix} 2 \\ 6 \\ -3 \end{pmatrix}$
 $E_{ABD} = g$
 $m=2 \quad (s=-7/11; t=12/11)$

c) 
 $\vec{AB} = \begin{pmatrix} -9 \\ 1 \\ -4 \end{pmatrix}$
 $\vec{AD} = \begin{pmatrix} 3 \\ -4 \\ -6 \end{pmatrix}$
 $\vec{DC} = \begin{pmatrix} -9 \\ 1 \\ -4 \end{pmatrix} = \vec{AB}$
 $\vec{BC} = \begin{pmatrix} 3 \\ -4 \\ -6 \end{pmatrix} = \vec{AD}$
 Parallelogram
 $\cos \alpha = \frac{\vec{AB} \cdot \vec{AD}}{|\vec{AB}| |\vec{AD}|} = \frac{-7}{\sqrt{98} \sqrt{61}} \Rightarrow \alpha = 95,19^\circ = \delta$
 $\beta = 180^\circ - \alpha = 84,8^\circ = \delta$

$A = |\vec{AB} \times \vec{AD}| = \left| \begin{pmatrix} -22 \\ -66 \\ 33 \end{pmatrix} \right| = 77$

3. A: 1..9
B: 1..5

a)



i) $P(gg) = \frac{4}{9} \cdot \frac{3}{8} + \frac{2}{5} \cdot \frac{1}{4} = \frac{4}{15} = 26,67\%$

ii) $P(A|gg) = \frac{\frac{4}{9} \cdot \frac{3}{8} + \frac{5}{9} \cdot \frac{2}{5}}{\frac{4}{9} \cdot \frac{3}{8} + \frac{5}{9} \cdot \frac{2}{5} + \frac{2}{5} \cdot \frac{1}{4} + \frac{2}{5} \cdot \frac{4}{5}} = \frac{15}{28} = 53,57\%$

iii) $P(uu) = \frac{5}{9} \cdot \frac{3}{5} + \frac{3}{5} \cdot \frac{1}{5} = \frac{2}{3} = 66,67\%$

b) $4P = (2,2); (\frac{1}{3}, 4); (4, \frac{1}{3})$
 $\frac{1}{9} \cdot \frac{1}{5} + 3 \cdot \frac{1}{9} \cdot \frac{1}{5} + 3 \cdot \frac{1}{9} \cdot \frac{1}{5} = \frac{7}{45}$

c) $2,4P: 1 - P(<4P)$
 $= 1 - \frac{7}{45}$
 $= \frac{38}{45} = 84,44\%$
 $3P: (\frac{1}{2}, 3); (3, \frac{1}{2})$
 $2 \cdot \frac{1}{9} \cdot \frac{1}{5} + 2 \cdot \frac{1}{9} \cdot \frac{1}{5} = \frac{4}{45}$
 $2P: (1, 4); (4, 1); (1, 1); (1, 1)$
 $\frac{3}{45}$

d) $P((x,y) | 6P)$
 $6P: (3, 3); (6, \frac{1}{3})$
 $= \frac{5}{6} = 83,33\%$

4.1. $V = 27 = G \cdot h = \frac{g^2}{4} \sqrt{3} \cdot h$; $\sigma = \frac{g^2}{4} \sqrt{3} + 3 \cdot s \cdot h \rightarrow \min$
 $h = \frac{108}{s^2 \sqrt{3}} \rightarrow$
 $= \frac{81}{4} s^2 + 3s \cdot \frac{108}{\sqrt{3} s^2}$
 $= \frac{\sqrt{3}}{4} s^2 + \sqrt{3} \cdot 108 \cdot \frac{1}{s}$
 $\sigma' = \frac{\sqrt{3}}{2} s - \sqrt{3} \cdot 108 \cdot \frac{1}{s^2}$
 $\sigma'' = \frac{\sqrt{3}}{1} + \sqrt{3} \cdot 216 \cdot \frac{1}{s^3} > 0 \text{ f. } s > 0$

$\sigma' = 0$
 $s = \sqrt[3]{216} = 6$
 $\sigma'' > 0 \Rightarrow \min$
 Runden 1): $\lim_{s \rightarrow 0} \sigma(s) = \infty$
 $\lim_{s \rightarrow \infty} \sigma(s) = \infty$ } $s = 6 \text{ Min}$
 $h = \sqrt{3}$

4.2. $f(x) = e^x (x^2 - \frac{7}{2}x + 2)$
 $f'(x) = e^x (x^2 - \frac{3}{2}x - \frac{3}{2})$
 $f''(x) = e^x (\underbrace{x^2 + \frac{1}{2}x - 3}_{=0}) = 0$
 $x_1 = -2$; $x_2 = \frac{3}{2}$
 $y_1 = 13e^{-2}$; $y_2 = -e^{-3/2}$

$d = \sqrt{(\frac{7}{2})^2 + (13e^{-2} + e^{-3/2})^2}$
 $= 7,155$